Method	GSM8K			Humaneval			MMLU-Pro		
	maj@2	maj@4	maj@8	pass@2	pass@4	pass@8	maj@2	maj@4	maj@8
Self-consistency	19.26 $\pm$0.36	25.32 $\pm$0.13	31.61 $\pm$0.12	18.29 $\pm$3.58	26.22 $\pm$3.14	33.54 $\pm$0.81	13.61 $\pm$0.11	15.15 $\pm$0.95	16.24 $\pm$0.91
Self-consistency (top-p&top-k)	25.47 $\pm$0.07	31.84 $\pm$0.13	38.74 $\pm$0.20	31.71 $\pm$3.18	36.59 $\pm$0.92	45.73 $\pm$0.58	14.96 $\pm$0.97	16.66 $\pm$0.91	17.35 $\pm$0.88
DLE (top-p&top-k)	**34.80** $\pm$0.42	**41.17** $\pm$0.71	**44.43** $\pm$0.53	**39.02** $\pm$0.88	**43.90** $\pm$0.66	**51.22** $\pm$0.38	**16.36** $\pm$0.92	17.30 $\pm$0.88	18.19 $\pm$0.85
Self-consistency (min-p)	26.46 $\pm$0.32	33.51 $\pm$0.56	39.04 $\pm$0.47	32.93 $\pm$0.12	38.41 $\pm$0.85	46.34 $\pm$0.55	14.99 $\pm$0.97	16.36 $\pm$0.92	17.26 $\pm$0.88
DLE (min-p)	**33.74** $\pm$0.56	**39.58** $\pm$0.31	**43.97** $\pm$0.65	**38.41** $\pm$0.91	**45.12** $\pm$0.61	**53.05** $\pm$0.31	**16.45** $\pm$0.92	17.40 $\pm$0.87	17.89 $\pm$0.86
Self-consistency (ε-sampling)	26.84 $\pm$0.01	33.51 $\pm$0.02	40.94 $\pm$0.06	32.93 $\pm$3.11	39.63 $\pm$0.81	47.56 $\pm$0.51	14.89 $\pm$0.97	16.07 $\pm$0.92	17.20 $\pm$0.89
DLE (ε-sampling)	**34.57** $\pm$0.28	**40.64** $\pm$0.17	**44.05** $\pm$0.11	**38.41** $\pm$0.90	**45.12** $\pm$0.61	**52.44** $\pm$0.34	16.64 $\pm$0.91	**17.37** $\pm$0.87	17.97 $\pm$0.86

Table 1: **DLE with various samplers improves performance compared to stochastic self-consistency.** Performance (maj@k and pass@k) of various methods on GSM8K, Humaneval and MMLU-Pro with the Qwen2.5-0.5B-Instruct model. Values are mean$\pm$std over three seeds. Statistically significant differences ($p < 0.05$) marked in **bold**.

Method	GSM8K			Humaneval			MMLU-Pro		
	maj@2	maj@4	maj@8	pass@2	pass@4	pass@8	maj@2	maj@4	maj@8
Self-consistency	78.77 ± 0.15	86.05 ± 0.12	89.16 ± 0.02	64.02 ± 0.31	77.44 ± 0.88	85.98 ± 0.42	45.79 ± 0.71	48.43 ± 0.66	51.75 ± 0.58
Self-consistency (top-p+top-k)	81.12 ± 0.01	87.79 ± 0.01	89.23 ± 0.01	68.90 ± 0.05	78.05 ± 0.76	87.20 ± 0.31	45.89 ± 0.68	47.11 ± 0.63	50.13 ± 0.61
DLE (top-p+top-k)	**84.23** ± 0.15	87.79 ± 0.06	89.69 ± 0.09	**81.10** ± 0.62	**85.37** ± 0.34	87.20 ± 0.28	46.78 ± 0.64	**48.61** ± 0.59	50.52 ± 0.57
Self-consistency (min-p)	81.58 ± 0.11	87.49 ± 0.21	89.84 ± 0.06	70.43 ± 0.97	80.38 ± 0.61	83.40 ± 0.55	46.14 ± 0.67	48.09 ± 0.62	49.88 ± 0.60
DLE (min-p)	**84.61** ± 0.25	**88.48** ± 0.02	**89.16** ± 0.04	**79.88** ± 0.68	**84.15** ± 0.41	**86.59** ± 0.30	46.84 ± 0.63	48.55 ± 0.58	50.40 ± 0.56
Self-consistency (ε-sampling)	81.43 ± 0.56	87.87 ± 0.44	89.92 ± 0.38	72.56 ± 0.91	81.10 ± 0.57	88.41 ± 0.22	46.19 ± 0.66	48.31 ± 0.61	51.89 ± 0.55
DLE (ε-sampling)-DivFirst	**84.91** ± 0.45	**88.48** ± 0.40	89.92 ± 0.37	**81.10** ± 0.58	**86.59** ± 0.29	87.80 ± 0.25	46.89 ± 0.62	48.90 ± 0.57	50.71 ± 0.55
DLE (ε-sampling)-ProbFirst	**84.15** ± 0.48	87.95 ± 0.42	88.86 ± 0.40	**79.88** ± 0.66	**84.76** ± 0.38	87.20 ± 0.27	46.88 ± 0.63	48.72 ± 0.58	50.62 ± 0.56
DLE (ε-sampling)-RandBranch	**83.78** ± 0.50	87.11 ± 0.45	88.78 ± 0.41	**76.83** ± 0.74	**83.54** ± 0.45	88.41 ± 0.23	46.70 ± 0.64	48.72 ± 0.59	50.69 ± 0.56

Table 2: Performance (maj@k and pass@k) of various methods on GSM8K, Humaneval and MMLU-Pro with the Qwen2.5-7B-Instruct model. Values are mean±std over three seeds. Statistically significant differences ($p < 0.05$) marked in **bold**.

Method	GSM8K			Humaneval			MMLU-Pro		
	maj@2	maj@4	maj@8	pass@2	pass@4	pass@8	maj@2	maj@4	maj@8
Self-consistency	88.48 $_{\pm 0.50}$	91.51 $_{\pm 0.49}$	92.49 $_{\pm 0.93}$	67.68 $_{\pm 0.28}$	78.05 $_{\pm 0.71}$	88.41 $_{\pm 0.18}$	52.01 $_{\pm 0.58}$	55.50 $_{\pm 0.52}$	58.39 $_{\pm 0.48}$
Self-consistency (top-p+top-k)	89.61 $_{\pm 0.31}$	91.21 $_{\pm 0.23}$	92.80 $_{\pm 0.91}$	72.56 $_{\pm 0.86}$	84.76 $_{\pm 0.33}$	89.02 $_{\pm 0.14}$	52.68 $_{\pm 0.56}$	55.31 $_{\pm 0.51}$	57.74 $_{\pm 0.49}$
DLE (top-p+top-k)	**89.92** $_{\pm 0.5}$	91.81 $_{\pm 0.74}$	91.74 $_{\pm 0.43}$	**76.83** $_{\pm 0.69}$	84.15 $_{\pm 0.36}$	88.41 $_{\pm 0.17}$	**54.50** $_{\pm 0.53}$	**56.27** $_{\pm 0.49}$	58.06 $_{\pm 0.47}$
Self-consistency (min-p)	89.99 $_{\pm 0.50}$	91.21 $_{\pm 0.92}$	92.34 $_{\pm 0.63}$	76.22 $_{\pm 0.71}$	81.10 $_{\pm 0.49}$	85.37 $_{\pm 0.34}$	52.68 $_{\pm 0.56}$	55.31 $_{\pm 0.51}$	57.74 $_{\pm 0.49}$
DLE (min-p)	90.52 $_{\pm 0.83}$	91.43 $_{\pm 0.84}$	91.74 $_{\pm 0.76}$	76.83 $_{\pm 0.68}$	**84.76** $_{\pm 0.31}$	**89.02** $_{\pm 0.13}$	**54.34** $_{\pm 0.53}$	56.13 $_{\pm 0.49}$	58.23 $_{\pm 0.47}$
Self-consistency (ε-sampling)	89.23 $_{\pm 0.72}$	91.36 $_{\pm 0.40}$	92.72 $_{\pm 0.50}$	73.17 $_{\pm 0.83}$	83.54 $_{\pm 0.39}$	88.41 $_{\pm 0.17}$	52.94 $_{\pm 0.55}$	56.48 $_{\pm 0.49}$	58.56 $_{\pm 0.46}$
DLE (ε-sampling)-DivFirst	90.60 $_{\pm 0.35}$	91.51 $_{\pm 0.32}$	92.34 $_{\pm 0.29}$	**76.22** $_{\pm 0.70}$	**85.98** $_{\pm 0.27}$	88.41 $_{\pm 0.16}$	**54.66** $_{\pm 0.52}$	56.40 $_{\pm 0.48}$	58.18 $_{\pm 0.46}$
DLE (ε-sampling)-ProbFirst	90.07 $_{\pm 0.37}$	91.43 $_{\pm 0.32}$	92.19 $_{\pm 0.30}$	**76.83** $_{\pm 0.67}$	**85.98** $_{\pm 0.26}$	89.02 $_{\pm 0.13}$	**54.33** $_{\pm 0.53}$	56.38 $_{\pm 0.49}$	58.30 $_{\pm 0.46}$
DLE (ε-sampling)-RandBranch	89.92 $_{\pm 0.38}$	91.51 $_{\pm 0.32}$	92.42 $_{\pm 0.29}$	**78.66** $_{\pm 0.61}$	**84.76** $_{\pm 0.31}$	88.41 $_{\pm 0.16}$	**54.88** $_{\pm 0.52}$	56.52 $_{\pm 0.48}$	58.29 $_{\pm 0.46}$

Table 3: Performance (maj@k and pass@k) of various methods on GSM8K, Humaneval and MMLU-Pro with the Qwen2.5-14B-Instruct model. Values are mean±std over three seeds. Statistically significant differences ($p < 0.05$) marked in **bold**.

Method	GSM8K			Humaneval			MMLU-Pro		
	maj@2	maj@4	maj@8	pass@2	pass@4	pass@8	maj@2	maj@4	maj@8
Self-consistency	18.50 ± 0.42	24.64 ± 0.31	32.60 ± 0.18	25.00 ± 3.21	37.20 ± 0.84	46.95 ± 0.49	13.73 ± 0.94	16.71 ± 0.88	19.02 ± 0.82
Self-consistency (top-p+top-k)	27.60 ± 0.28	33.21 ± 0.19	39.65 ± 0.09	31.71 ± 0.97	44.51 ± 0.61	54.88 ± 0.28	17.30 ± 0.88	19.30 ± 0.83	20.78 ± 0.79
DLE (top-p+top-k)	**33.06** ± 0.15	**36.01** ± 0.48	38.06 ± 0.34	**42.07** ± 0.58	**53.66** ± 0.29	**59.15** ± 0.11	**20.62** ± 0.81	**22.17** ± 0.77	**22.86** ± 0.74
Self-consistency (min-p)	30.10 ± 0.21	35.25 ± 0.11	**40.25** ± 0.15	37.20 ± 0.74	46.34 ± 0.46	54.88 ± 0.27	17.73 ± 0.87	20.09 ± 0.81	21.29 ± 0.78
DLE (min-p)	**33.21** ± 0.14	**36.01** ± 0.18	38.44 ± 0.13	**42.07** ± 0.57	**53.05** ± 0.31	**59.76** ± 0.08	**20.71** ± 0.81	**22.17** ± 0.77	22.85 ± 0.74
Self-consistency (ε-sampling)	27.75 ± 0.27	33.89 ± 0.17	39.95 ± 0.17	37.80 ± 0.71	48.78 ± 0.41	54.88 ± 0.28	18.87 ± 0.85	20.69 ± 0.80	21.82 ± 0.77
DLE (ε-sampling)-DivFirst	**31.39** ± 0.18	**34.65** ± 0.13	36.85 ± 0.16	31.71 ± 0.96	39.63 ± 0.79	46.95 ± 0.50	19.76 ± 0.83	21.23 ± 0.79	22.15 ± 0.76
DLE (ε-sampling)-ProbFirst	**33.21** ± 0.14	**36.01** ± 0.18	38.06 ± 0.14	**41.46** ± 0.60	**52.44** ± 0.33	**59.76** ± 0.08	19.61 ± 0.83	21.13 ± 0.79	22.95 ± 0.74
DLE (ε-sampling)-RandBranch	**33.13** ± 0.14	34.80 ± 0.12	38.44 ± 0.13	**43.29** ± 0.55	48.17 ± 0.43	**57.32** ± 0.18	19.75 ± 0.83	20.90 ± 0.80	21.97 ± 0.76

Table 4: Performance (maj@k and pass@k) of various methods on GSM8K, Humaneval and MMLU-Pro with the Llama3.2-1B-Instruct model. Values are mean±std over three seeds. Statistically significant differences ($p < 0.05$) marked in **bold**.

Method	GSM8K			Humaneval			MMLU-Pro		
	maj@2	maj@4	maj@8	pass@2	pass@4	pass@8	maj@2	maj@4	maj@8
Self-consistency	43.29 ± 0.01	54.36 ± 0.01	65.58 ± 0.01	46.34 ± 0.49	57.93 ± 0.18	70.12 ± 0.79	24.88 ± 0.74	29.31 ± 0.68	33.04 ± 0.63
Self-consistency (top-p+top-k)	56.25 ± 0.01	65.81 ± 0.19	**72.48** ± 0.10	49.39 ± 0.39	64.02 ± 0.96	73.78 ± 0.67	29.74 ± 0.68	33.64 ± 0.63	36.17 ± 0.60
DLE (top-p+top-k)	**64.52** ± 0.06	**69.45** ± 0.04	71.49 ± 0.10	**59.15** ± 0.10	**67.07** ± 0.85	**76.83** ± 0.55	**33.22** ± 0.63	**35.73** ± 0.59	**37.28** ± 0.57
Self-consistency (min-p)	59.89 ± 0.24	68.39 ± 0.33	**73.69** ± 0.53	51.83 ± 0.32	65.85 ± 0.91	76.83 ± 0.54	30.68 ± 0.67	34.73 ± 0.61	36.08 ± 0.60
DLE (min-p)	**64.44** ± 0.03	68.84 ± 0.18	71.57 ± 0.06	**59.15** ± 0.09	**68.29** ± 0.81	76.83 ± 0.55	**33.23** ± 0.63	35.71 ± 0.59	36.91 ± 0.58
Self-consistency (ε-sampling)	58.38 ± 0.88	62.33 ± 0.83	69.14 ± 0.74	55.49 ± 0.21	64.46 ± 0.94	75.61 ± 0.59	30.98 ± 0.66	35.37 ± 0.60	37.36 ± 0.57
DLE (ε-sampling)-DivFirst	**62.62** ± 0.83	**67.40** ± 0.77	69.52 ± 0.74	55.49 ± 0.20	65.24 ± 0.90	71.95 ± 0.72	**33.19** ± 0.63	35.65 ± 0.59	37.23 ± 0.57
DLE (ε-sampling)-ProbFirst	**64.44** ± 0.81	**68.92** ± 0.75	**71.65** ± 0.72	**59.15** ± 0.08	**69.51** ± 0.78	**78.66** ± 0.49	**33.18** ± 0.63	35.88 ± 0.58	37.53 ± 0.56
DLE (ε-sampling)-RandBranch	**64.52** ± 0.81	**68.31** ± 0.76	**72.33** ± 0.71	**57.93** ± 0.14	65.24 ± 0.90	73.17 ± 0.69	**33.22** ± 0.63	35.60 ± 0.59	37.49 ± 0.56

Table 5: Performance (maj@k and pass@k) of various methods on GSM8K, Humaneval and MMLU-Pro with the Llama3.2-3B-Instruct model. Values are mean±std over three seeds. Statistically significant differences ($p < 0.05$) marked in **bold**.